Homework 7: Linear Systems Solutions

Math 124, Fall 2026 at the University of Michigan

due Sunday, November 8th at 11:59PM

Write your solutions to the following problems either by writing them on a piece of paper or on a tablet and scanning your answers as a PDF. Note that you are not allowed to use LaTeX, Google Docs, or any other digital document creation software to type your answers. Homeworks are due to Pensive by 11:59PM on the due date. See the syllabus for details on the slip day policy.

Homework will be evaluated not only on the correctness of your answers, but on your ability to present your ideas clearly and logically. You should always explain and justify your conclusions, using sound reasoning. Your goal should be to convince the reader of your assertions. If a question does not require explanation, it will be explicitly stated.

Before proceeding, make sure you're familiar with the collaboration policy.

Total Points: $6 + 15 + 20 + 15 + 26 + 18 = 100$

Problem 1: Homework 6 Solutions Review (6 pts)

Review the solutions to Homework 6. Pick **two problem parts** (for example, Problem 3a and Problem 5b) from Homework 6 in which your solutions have the most room for improvement, i.e., where they have unsound reasoning, could be significantly more efficient or clearer, etc. **Include a screenshot of your solution to each problem part**, and in a few sentences, explain what was deficient and how it could be fixed.

Alternatively, if you think one of your solutions is significantly better than the posted one, copy it here and explain why you think it is better. If you didn't do Homework 6, choose two problem parts from it that look challenging to you, and in a few sentences, explain the key ideas behind their solutions in your own words.

Solution:

Problem 2: When a Pivot Goes Missing (15 pts)

Let

$$A = \begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & 0 \\ 3 & -6 & 1 \end{bmatrix}.$$

a) (7 pts) Find a factorization $A = LU$, where L is lower triangular with ones on its diagonal and U is in row echelon form. Identify the pivot columns of U.
Hint: Don't forget about the requirement that U is in row echelon form!

Solution: We start by writing the entries we need to find explicitly:

$$\underbrace{\begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & 0 \\ 3 & -6 & 1 \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ \ell_{21} & 1 & 0 \\ \ell_{31} & \ell_{32} & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}}_{U}.$$

The zeros below the diagonal of U are required by echelon form. We will also check that its pivots move strictly to the right and that its zero rows come last.
Step 1: Find the first row of U. The first row of L is $\begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$, so the first row of LU is exactly the first row of U. Matching it with the first row of A gives

$$u_{11} = 1, \qquad u_{12} = -2, \qquad u_{13} = 1.$$

Step 2: Find the first column of L. Matching entries $(2,1)$ and $(3,1)$ gives

$$\ell_{21}(1) = -2 \implies \ell_{21} = -2, \qquad \ell_{31}(1) = 3 \implies \ell_{31} = 3.$$

Our partially completed factorization is now

$$A = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & \ell_{32} & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 & 1 \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}.$$

Step 3: Find the second row of U. The second row of LU is -2 times the first row of U, plus the second row of U. Matching entries $(2,2)$ and $(2,3)$ therefore gives

$$\begin{aligned} (-2)(-2) + u_{22} = 4 \quad &\implies \quad u_{22} = 0, \\ (-2)(1) + u_{23} = 0 \quad &\implies \quad u_{23} = 2. \end{aligned}$$

The second row is $\begin{bmatrix} 0 & 0 & 2 \end{bmatrix}$, with its pivot in column 3. There is no column to its right, so echelon form forces the third row to be zero: $u_{33} = 0$.
Step 4: Find ℓ_{32} using the pivot in column 3. First check the skipped column 2: entry $(3,2)$ gives $3(-2) + \ell_{32}(0) = -6$, which already holds. It does not determine ℓ_{32}. Entry $(3,3)$ gives

$$3(1) + \ell_{32}(2) + 0 = 1 \quad \implies \quad \ell_{32} = -1.$$

Thus

$$\boxed{A = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & -1 & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} 1 & -2 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}}_{U}.}$$

Multiplication verifies the last two rows: they are $\begin{bmatrix} -2 & 4 & -2+2 \end{bmatrix}$ and $\begin{bmatrix} 3 & -6 & 3-2 \end{bmatrix}$, exactly as in A. The pivot columns of U are **1 and 3**.

b) (8 pts) Use your factors to find all solutions of each system:

$$A\vec{x} = \begin{bmatrix} 3 \\ -2 \\ 5 \end{bmatrix}, \qquad A\vec{x} = \begin{bmatrix} 3 \\ -2 \\ 6 \end{bmatrix}.$$

If there are infinitely many solutions, give a vector-parametric description of the solution set. If there is no solution, explain why.

Solution: Once we know $A = LU$, the system $A\vec{x} = \vec{b}$ becomes

$$L(U\vec{x}) = \vec{b}.$$

Introduce the intermediate vector $\vec{y} = U\vec{x}$. We can now solve two simpler systems in order:

$$\text{First solve } L\vec{y} = \vec{b}, \qquad \text{then solve } U\vec{x} = \vec{y}.$$

First system: $\vec{b} = \begin{bmatrix} 3 \\ -2 \\ 5 \end{bmatrix}$.

Step 1: Solve for $\vec{y}$ by forward substitution. Using the factors from part a), write

$$\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & -1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \\ 5 \end{bmatrix}.$$

Read the equations from top to bottom:

$$\begin{aligned} y_1 &= 3, \\ -2y_1 + y_2 = -2 \quad &\Longrightarrow \quad -6 + y_2 = -2 \quad \Longrightarrow \quad y_2 = 4, \\ 3y_1 - y_2 + y_3 = 5 \quad &\Longrightarrow \quad 9 - 4 + y_3 = 5 \quad \Longrightarrow \quad y_3 = 0. \end{aligned}$$

Thus $\vec{y} = \begin{bmatrix} 3 \\ 4 \\ 0 \end{bmatrix}$. This is the intermediate vector, not yet the solution $\vec{x}$.

Step 2: Solve for $\vec{x}$ in $U\vec{x} = \vec{y}$ using back substitution. The second system is

$$\begin{bmatrix} 1 & -2 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \\ 0 \end{bmatrix},$$

which gives

$$x_1 - 2x_2 + x_3 = 3, \qquad 2x_3 = 4, \qquad 0 = 0.$$

The zero row gives a true equation, so the system is consistent. The pivot columns are 1 and 3, so x_1, x_3 are pivot variables and x_2 is free. Set $x_2 = t$, where t can be any real number. Starting with the last nonzero equation gives

$$2x_3 = 4 \quad \Longrightarrow \quad x_3 = 2.$$

Substitute $x_2 = t$ and $x_3 = 2$ into the first equation:

$$x_1 - 2t + 2 = 3 \quad \Longrightarrow \quad x_1 = 1 + 2t.$$

Therefore

$$\boxed{\vec{x} = \begin{bmatrix} 1 + 2t \\ t \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + t \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \qquad t \in \mathbb{R}.}$$

There are infinitely many solutions $\vec{x}$, one for each value of $t \in \mathbb{R}$.

Second system: $\vec{b} = \begin{bmatrix} 3 \\ -2 \\ 6 \end{bmatrix}$**.** The coefficient matrix has not changed, so we reuse the same L and U. Only the right-hand side in the first triangular solve changes:

$$\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & -1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \\ 6 \end{bmatrix}.$$

The first two equations are unchanged, giving $y_1 = 3$ and $y_2 = 4$. The last equation now gives

$$3(3) - 4 + y_3 = 6 \quad \Longrightarrow \quad y_3 = 1.$$

Thus the second triangular solve would have to satisfy

$$\begin{bmatrix} 1 & -2 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \\ 1 \end{bmatrix}.$$

Its last row says $0 = 1$, which is impossible. No choices of x_1, x_2, x_3 can satisfy this equation, even though column 2 has no pivot. Therefore the second system has **no solution**.

Problem 3: Factoring a Rectangular Matrix (20 pts)

Let

$$B = \begin{bmatrix} 2 & -2 & 1 & 3 \\ 6 & -6 & 3 & 9 \\ -2 & 2 & 3 & 1 \end{bmatrix}.$$

a) (5 pts) Attempt to find a factorization $B = LU$ without reordering the rows, where L has ones on its diagonal and U is in row echelon form. Explain why a row exchange is necessary.

Solution: Without changing the row order, the required shapes give

$$\underbrace{\begin{bmatrix} 2 & -2 & 1 & 3 \\ 6 & -6 & 3 & 9 \\ -2 & 2 & 3 & 1 \end{bmatrix}}_{B} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ \ell_{21} & 1 & 0 \\ \ell_{31} & \ell_{32} & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} u_{11} & u_{12} & u_{13} & u_{14} \\ 0 & u_{22} & u_{23} & u_{24} \\ 0 & 0 & u_{33} & u_{34} \end{bmatrix}}_{U}.$$

First row and first column. The first row of LU equals the first row of U, so it must be $\begin{bmatrix} 2 & -2 & 1 & 3 \end{bmatrix}$. Matching the first column then gives

$$2\ell_{21} = 6 \implies \ell_{21} = 3, \qquad 2\ell_{31} = -2 \implies \ell_{31} = -1.$$

Second row. Match the remaining entries of row 2:

$$\begin{aligned} 3(-2) + u_{22} = -6 &\implies u_{22} = 0, \\ 3(1) + u_{23} = 3 &\implies u_{23} = 0, \\ 3(3) + u_{24} = 9 &\implies u_{24} = 0. \end{aligned}$$

Thus row 2 of U is entirely zero. Echelon form requires every row below it to be zero too, forcing $u_{33} = u_{34} = 0$. The only possible product would therefore have the form

$$LU = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ -1 & \ell_{32} & 1 \end{bmatrix} \begin{bmatrix} 2 & -2 & 1 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Its third row is $\begin{bmatrix} -2 & 2 & -1 & -3 \end{bmatrix}$, whereas the third row of B is $\begin{bmatrix} -2 & 2 & 3 & 1 \end{bmatrix}$. No value of ℓ_{32} can fix this, because it multiplies a zero row of U. Hence **a row exchange is necessary**.

b) (8 pts) Find a permutation matrix P and factors L, U such that $PB = LU$, with L square and lower triangular with ones on its diagonal and U in row echelon form. Identify the pivot columns of U, and verify your factorization by multiplying LU and showing it is equal to PB.

Solution: Swap rows 2 and 3 using

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

We now match entries in

$$\underbrace{\begin{bmatrix} 2 & -2 & 1 & 3 \\ -2 & 2 & 3 & 1 \\ 6 & -6 & 3 & 9 \end{bmatrix}}_{PB} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ \ell_{21} & 1 & 0 \\ \ell_{31} & \ell_{32} & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} u_{11} & u_{12} & u_{13} & u_{14} \\ 0 & u_{22} & u_{23} & u_{24} \\ 0 & 0 & u_{33} & u_{34} \end{bmatrix}}_{U}.$$

Step 1: First row of U, then first column of L. The first row of U is $\begin{bmatrix} 2 & -2 & 1 & 3 \end{bmatrix}$. The first column gives

$$2\ell_{21} = -2 \implies \ell_{21} = -1, \qquad 2\ell_{31} = 6 \implies \ell_{31} = 3.$$

Step 2: Second row of U. Matching row 2 gives

$$\begin{aligned} (-1)(-2) + u_{22} = 2 &\implies u_{22} = 0, \\ (-1)(1) + u_{23} = 3 &\implies u_{23} = 4, \\ (-1)(3) + u_{24} = 1 &\implies u_{24} = 4. \end{aligned}$$

The second pivot is in column 3. The next pivot, if any, must be to its right, so $u_{33} = 0$. At this stage,

$$PB = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 3 & \ell_{32} & 1 \end{bmatrix} \begin{bmatrix} 2 & -2 & 1 & 3 \\ 0 & 0 & 4 & 4 \\ 0 & 0 & 0 & u_{34} \end{bmatrix}.$$

Step 3: Second column of L, then the last entry of U. The skipped column 2 checks: $3(-2) + \ell_{32}(0) = -6$. Use the pivot in column 3 to find ℓ_{32}, then match column 4:

$$\begin{aligned} 3(1) + 4\ell_{32} = 3 &\implies \ell_{32} = 0, \\ 3(3) + 0(4) + u_{34} = 9 &\implies u_{34} = 0. \end{aligned}$$

Therefore

$$\boxed{PB = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} 2 & -2 & 1 & 3 \\ 0 & 0 & 4 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}}_{U}.}$$

Verification by multiplication gives

$$LU = \begin{bmatrix} 2 & -2 & 1 & 3 \\ -2 & 2 & -1+4 & -3+4 \\ 6 & -6 & 3 & 9 \end{bmatrix} = PB.$$

The pivot columns of U are **1 and 3**.

c) (7 pts) Use your factors to find all solutions of

$$B\vec{x} = \begin{bmatrix} 6 \\ 18 \\ 2 \end{bmatrix}.$$

Give a vector-parametric description of the solution set.

Solution: First reorder the right-hand side:

$$P\vec{b} = \begin{bmatrix} 6 \\ 2 \\ 18 \end{bmatrix}.$$

Forward substitution gives $y_1 = 6$, $-y_1 + y_2 = 2$, and $3y_1 + y_3 = 18$, so $\vec{y} = \begin{bmatrix} 6 \\ 8 \\ 0 \end{bmatrix}$. The remaining equations are

$$2x_1 - 2x_2 + x_3 + 3x_4 = 6, \qquad 4x_3 + 4x_4 = 8, \qquad 0 = 0.$$

The free variables are x_2 and x_4. Set $x_2 = s$, $x_4 = t$. Then $x_3 = 2 - t$ and $x_1 = 2 + s - t$, giving

$$\boxed{\vec{x} = \begin{bmatrix} 2 \\ 0 \\ 2 \\ 0 \end{bmatrix} + s \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ -1 \\ 1 \end{bmatrix}, \qquad s, t \in \mathbb{R}.}$$

The zero row is consistent, and each choice of s, t uniquely determines the pivot variables, so these are all solutions.

Problem 4: Working Backward (15 pts)

For each part, give a matrix U in row echelon form and a vector $\vec{y}$ such that the system $U\vec{x} = \vec{y}$ has the stated properties. Write your system as a matrix equation, and give a vector-parametric description of the solution set, or explain why it has no solution. Each part may use a different matrix.

a) (3 pts) U is 3×3, and the system has a unique solution.

Solution: One example is

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.$$

Every variable is a pivot variable, and the unique solution is $\vec{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$.

b) (3 pts) U is 3×3, and the solution set is a line in $\mathbb{R}^3$ that does not pass through the origin.

Solution: One example is

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

The equations fix $x_1 = 1$, $x_2 = 0$, while $x_3 = t$ is free:

$$\vec{x} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \qquad t \in \mathbb{R}.$$

This is a line in $\mathbb{R}^3$. It does not contain the origin because every solution has first coordinate 1.

c) (3 pts) U is 3×4, and the solution set is a plane through the origin in $\mathbb{R}^4$.

Solution: One example is

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Here $x_1 = x_2 = 0$, while $x_3 = s$ and $x_4 = t$ can be chosen freely:

$$\vec{x} = s \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \qquad s, t \in \mathbb{R}.$$

The two parameters vary the third and fourth coordinates independently, giving a plane in $\mathbb{R}^4$. Taking $s = t = 0$ gives the origin.

d) (3 pts) U is 4×3, and the system has no solution.

Solution: One example is

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

The fourth equation is $0 = 1$, so there is no solution.

e) (3 pts) U is 4×2, and the system has a unique solution.

Solution: One example is

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 0 \\ 0 \end{bmatrix}.$$

Both zero rows give $0 = 0$, so the system is consistent. Both columns have pivots, so there are no free variables. The unique solution is $\vec{x} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$.

Problem 5: A Plane of Possibilities (26 pts)

Suppose $\vec{v} = \begin{bmatrix}1\\1\\1\end{bmatrix}$ and $\vec{w} = \begin{bmatrix}1\\2\\3\end{bmatrix}$, and let C be the 3×2 matrix whose columns are $\vec{v}$ and $\vec{w}$:

$$C = \begin{bmatrix}| & |\\ \vec{v} & \vec{w}\\ | & |\end{bmatrix} = \begin{bmatrix}1 & 1\\1 & 2\\1 & 3\end{bmatrix}.$$

A possible output of a matrix is any vector that can be produced by multiplying that matrix by a vector. For example, $\begin{bmatrix}5\\9\\13\end{bmatrix}$ is a possible output of C, since

$$\begin{bmatrix}5\\9\\13\end{bmatrix} = C\begin{bmatrix}1\\4\end{bmatrix}.$$

a) (4 pts) Find a factorization $C = LU$, where L is square and lower triangular with ones on its diagonal and U is in row echelon form.

Solution: Here C has shape 3×2, so L must be 3×3 and U must be 3×2. An echelon matrix with only two columns can have at most two nonzero rows, so its third row must be zero. Start with

$$\underbrace{\begin{bmatrix}1 & 1\\1 & 2\\1 & 3\end{bmatrix}}_{C} = \underbrace{\begin{bmatrix}1 & 0 & 0\\ \ell_{21} & 1 & 0\\ \ell_{31} & \ell_{32} & 1\end{bmatrix}}_{L}\underbrace{\begin{bmatrix}u_{11} & u_{12}\\0 & u_{22}\\0 & 0\end{bmatrix}}_{U}.$$

Step 1: First row of U. Matching the first row gives $u_{11} = 1$ and $u_{12} = 1$.
Step 2: First column of L. Matching the first column gives

$$\ell_{21}(1) = 1 \implies \ell_{21} = 1, \qquad \ell_{31}(1) = 1 \implies \ell_{31} = 1.$$

We have now determined

$$C = \begin{bmatrix}1 & 0 & 0\\1 & 1 & 0\\1 & \ell_{32} & 1\end{bmatrix}\begin{bmatrix}1 & 1\\0 & u_{22}\\0 & 0\end{bmatrix}.$$

Step 3: Second row of U. Entry $(2,2)$ gives

$$1(1) + u_{22} = 2 \quad \implies \quad u_{22} = 1.$$

Step 4: Second column of L. Entry $(3,2)$ gives

$$1(1) + \ell_{32}(1) + 0 = 3 \quad \implies \quad \ell_{32} = 2.$$

Thus

$$\boxed{C = \underbrace{\begin{bmatrix}1 & 0 & 0\\1 & 1 & 0\\1 & 2 & 1\end{bmatrix}}_{L}\underbrace{\begin{bmatrix}1 & 1\\0 & 1\\0 & 0\end{bmatrix}}_{U}.}$$

The product is $LU = \begin{bmatrix}1 & 1\\1 & 1+1\\1 & 1+2\end{bmatrix} = C$. The two nonzero rows of U have pivots in columns 1 and 2, and its zero row comes last, as required.

b) (6 pts) What are all possible outputs of C? Using your factorization $C = LU$, determine the conditions on x, y, and z such that

$$C\vec{u} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

has a solution $\vec{u}$. The condition you find should be a single equation involving x, y, and z.

Solution: From part a), the factors in $C = LU$ are

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 2 & 1 \end{bmatrix}, \qquad U = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

To solve $C\vec{u} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, first solve $L\vec{t} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$:

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} t_1 \\ t_2 \\ t_3 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}.$$

Forward substitution gives

$$\begin{aligned} t_1 &= x, \\ t_1 + t_2 = y \quad &\Longrightarrow \quad t_2 = y - x, \\ t_1 + 2t_2 + t_3 = z \quad &\Longrightarrow \quad x + 2(y - x) + t_3 = z \\ &\Longrightarrow \quad t_3 = z + x - 2y. \end{aligned}$$

Thus $\vec{t} = \begin{bmatrix} x \\ y - x \\ z + x - 2y \end{bmatrix}$. Next, solve $U\vec{u} = \vec{t}$:

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} x \\ y - x \\ z + x - 2y \end{bmatrix}.$$

The equations are

$$u_1 + u_2 = x, \qquad u_2 = y - x, \qquad 0 = z + x - 2y.$$

The last equation can hold only if $z + x - 2y = 0$. When it does, the second equation determines u_2, and the first determines u_1. Therefore the system is consistent exactly when

$$\boxed{x - 2y + z = 0.}$$

c) (4 pts) Under the condition you found in part b), find the solution for the vector $\vec{u} \in \mathbb{R}^2$ in terms of x, y, and z. Is the solution unique?

Solution: Suppose $x - 2y + z = 0$. Part b) then gives $\vec{t} = \begin{bmatrix} x \\ y - x \\ 0 \end{bmatrix}$. Using U from part a), solve $U\vec{u} = \vec{t}$:

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} x \\ y - x \\ 0 \end{bmatrix}.$$

The last row gives $0 = 0$. The second row gives $u_2 = y - x$. Substitute this into the first row:

$$u_1 + u_2 = x \quad \Longrightarrow \quad u_1 + (y - x) = x \quad \Longrightarrow \quad u_1 = 2x - y.$$

Thus

$$\boxed{\vec{u} = \begin{bmatrix} 2x - y \\ y - x \end{bmatrix}}.$$

Both columns of U have pivots, so there are no free variables: the solution is unique whenever it exists.

d) (8 pts) Using your understanding of geometry and span from earlier in the semester, find the equation of the plane spanned by $\vec{v}$ and $\vec{w}$. Why do you get the same answer as in part b)?

Solution: The combinations are

$$s\vec{v} + t\vec{w} = s \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + t \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} s + t \\ s + 2t \\ s + 3t \end{bmatrix}.$$

Their coordinates satisfy $(s+t) - 2(s+2t) + (s+3t) = 0$. Conversely, if $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ satisfies $x - 2y + z = 0$, choose $s = 2x - y$ and $t = y - x$. Then

$$s\vec{v} + t\vec{w} = \begin{bmatrix} x \\ y \\ 2y - x \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}.$$

Thus the set is exactly the plane $x - 2y + z = 0$ through the origin in $\mathbb{R}^3$, with normal vector $\begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$.

e) (4 pts) Consider the matrix

$$F = \begin{bmatrix} | & | & | \\ \vec{v} & \vec{w} & \vec{v}+\vec{w} \\ | & | & | \end{bmatrix} = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 1 & 3 & 4 \end{bmatrix}.$$

Does F have the same possible outputs as C? Explain. Then, find two different inputs that F sends to

$$2\vec{v} + 3\vec{w} = \begin{bmatrix} 5 \\ 8 \\ 11 \end{bmatrix}.$$

Solution: Yes. For any input $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$,

$$F \begin{bmatrix} a \\ b \\ c \end{bmatrix} = a\vec{v} + b\vec{w} + c(\vec{v}+\vec{w}) = (a+c)\vec{v} + (b+c)\vec{w} = C \begin{bmatrix} a+c \\ b+c \end{bmatrix}.$$

So every output of F is an output of C. Conversely, $C \begin{bmatrix} s \\ t \end{bmatrix} = F \begin{bmatrix} s \\ t \\ 0 \end{bmatrix}$, so every output of C is an output of F. Two different inputs producing $2\vec{v} + 3\vec{w} = \begin{bmatrix} 5 \\ 8 \\ 11 \end{bmatrix}$ are

$$\boxed{\begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}}.$$

Problem 6: The Score Adds Up (18 pts)

Before proceeding, open the supplemental Jupyter Notebook. To open the notebook for Homework 7, click **this link**. Instructions on how to use Google Colab are at math124.org/running-code.

Complete Problem 6 in the notebook. Tasks 1–8 are worth $1+1+2+5+1+2+1+5=18$ points, respectively. Follow its orange instructions for what to include in your PDF submission to Pensive.

Solution: a) The first row gives

$$\boxed{32w_1 + 4w_2 + 7w_3 = 75.}$$

The vector $\vec{w}$ has shape 3×1. Since A has shape 660×3, $A\vec{w}$ has shape 660×1; each entry is the score predicted for one team in one game. There are **three unknown coefficients**, shared by all 660 equations.

Solution: b) Complete the factorization and the two triangular solves as follows:

```
Q, L, U = lu(S)
P = Q.T
y = solve_triangular(L, P @ b_small, lower=True)
w = solve_triangular(U, y, lower=False)
```

The selected rows give

$$S = \begin{bmatrix} 32 & 4 & 7 \\ 17 & 5 & 26 \\ 31 & 15 & 14 \end{bmatrix}, \qquad \vec{b}_{\text{small}} = \begin{bmatrix} 75 \\ 65 \\ 91 \end{bmatrix}.$$

The output, displayed to four decimal places, is

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \qquad L \approx \begin{bmatrix} 1 & 0 & 0 \\ 0.9688 & 1 & 0 \\ 0.5312 & 0.2584 & 1 \end{bmatrix},$$

$$U \approx \begin{bmatrix} 32 & 4 & 7 \\ 0 & 11.1250 & 7.2188 \\ 0 & 0 & 20.4157 \end{bmatrix}, \qquad \vec{y} \approx \begin{bmatrix} 75 \\ 18.3438 \\ 20.4157 \end{bmatrix}, \qquad \boxed{\vec{w} = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}}.$$

Keep the unrounded arrays when computing. The supplied checks produce

```
Factorization verified: True
Lower triangular system verified: True
Upper triangular system verified: True
Our rule matches the three selected scores: True
```

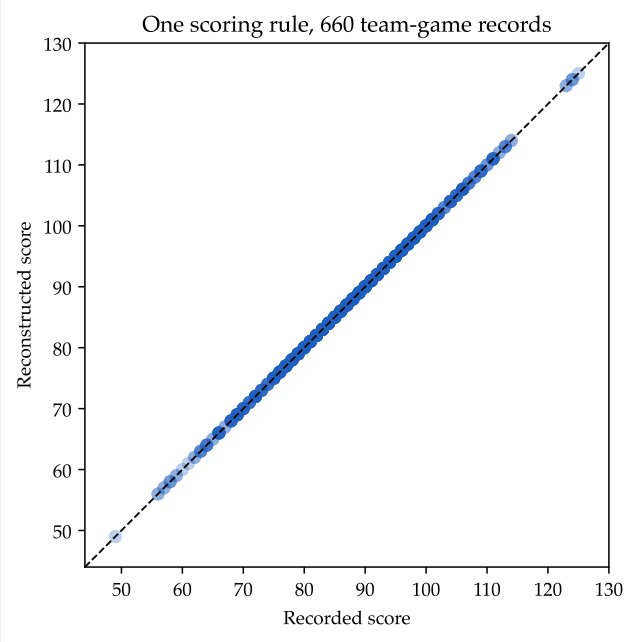
All three diagonal entries of U are nonzero. Thus there is a pivot in every column, and the two triangular solves determine exactly one coefficient vector. In particular, S is invertible, and no different coefficient vector can match these same three scores.

Solution: **c)** The completed line is

```
reconstructed_scores = A @ w
```

It has shape `(660, 1)`. Running the supplied comparison cell gives

```
Matching scores: 660 out of 660
All recorded scores reproduced: True
```



All plotted points lie on the line where the reconstructed and recorded scores agree.
The coefficients give the rule

$$\boxed{\text{score} = 2(\text{FGM}) + 3\text{PM} + \text{FTM}.}$$

Each made field goal contributes two points through `FGM`. A three-pointer is already included in `FGM`, so its `3PM` entry adds one more point, giving three in total. Each free throw contributes one point.

The system $A\vec{w} = \vec{b}$ has **exactly one solution**. The check over all 660 rows shows that $\begin{bmatrix}2\\1\\1\end{bmatrix}$ is a solution.
Any solution of the full system must also solve the three selected equations, whose solution is unique by part b).
With $\vec{u}, \vec{v}, \vec{z}$ denoting the columns of A, matrix-vector multiplication gives

$$\boxed{\vec{b} = 2\vec{u} + \vec{v} + \vec{z}.}$$

Solution: d) Two choices are

```
w_one = np.array([[2], [1], [1], [0]])
w_two = np.array([[0], [3], [1], [2]])
```

The first uses the rule from part c). The second counts three points for each three-pointer, one for each free throw, and two for each two-pointer. The supplied checks give

```
First vector reproduces every score: True
Second vector reproduces every score: True
The coefficient vectors are different: True
```

Complete the direction cell with

```
direction = w_two - w_one
```

Thus

$$\vec{d} = \begin{bmatrix} -2 \\ 2 \\ 0 \\ 2 \end{bmatrix}.$$

The checks produce

```
The difference is nonzero: True
C sends the difference to zero: True
t = -3: all scores match? True
t = 0: all scores match? True
t = 0.5: all scores match? True
t = 2: all scores match? True
```

To establish the result for every real t, first note that

$$C\vec{d} = C(\vec{w}_{\text{two}} - \vec{w}_{\text{one}}) = C\vec{w}_{\text{two}} - C\vec{w}_{\text{one}} = \vec{b} - \vec{b} = \vec{0}.$$

Distributing matrix multiplication then gives

$$\boxed{C(\vec{w}_{\text{one}} + t\vec{d}) = C\vec{w}_{\text{one}} + tC\vec{d} = \vec{b} \quad \text{for every } t \in \mathbb{R}.}$$

Different values of t give different fourth coordinates, namely $2t$. There are therefore infinitely many solutions.
Adding `2PM` did **not** change the possible outputs. The columns of C are $\vec{u}, \vec{v}, \vec{z}, \vec{u} - \vec{v}$, so

$$C \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = (a+d)\vec{u} + (b-d)\vec{v} + c\vec{z} = A \begin{bmatrix} a+d \\ b-d \\ c \end{bmatrix}.$$

Thus every output of C is an output of A. Conversely, $A \begin{bmatrix} a \\ b \\ c \end{bmatrix} = C \begin{bmatrix} a \\ b \\ c \\ 0 \end{bmatrix}$, so every output of A is an output of C.

Solution: **e)** Complete the comparison with

```
changed_matches = np.isclose(
    (A @ w)[:, 0], b_changed[:, 0]
)
```

The output is

```
Matching scores after the change: 659 out of 660
Rows that do not match: [137]
```

Row 137 describes the Los Angeles Sparks against the Las Vegas Aces on June 2. Its statistics are 20 field goals, 6 three-pointers, and 23 free throws. The original score is

$$2(20) + 6 + 23 = 69,$$

while the experiment changes it to 70.

1. No coefficient vector can satisfy all the altered equations. The three selected rows (0, 2, and 4) have not changed. Any candidate must therefore still solve $S\vec{w} = \vec{b}_{\text{small}}$. Because S is invertible, those three equations force $\vec{w} = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$. But this vector predicts 69 in row 137, whereas the altered equation requires 70. This is a contradiction. Hence $A\vec{w} = \vec{b}_{\text{changed}}$ has **no solution**.

2. Using all four columns cannot rescue the altered scores. Part d) showed that every output of C is also an output of A. If some four-entry vector produced $\vec{b}_{\text{changed}}$ through C, the corresponding three-entry vector would produce it through A, contradicting part 1. Thus the four-column system also has **no solution**.